3.

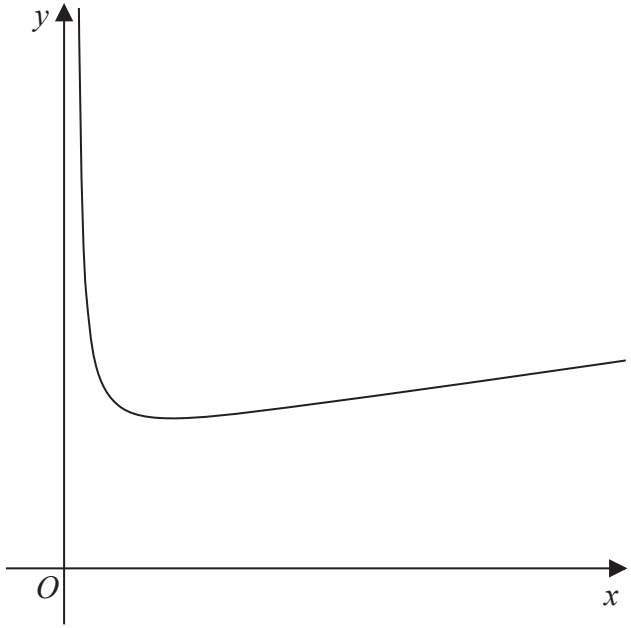


Figure 1

Figure 1 shows a sketch of a curve with equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{2x + 3}{\sqrt{4x - 1}} \qquad x > \frac{1}{4}$$

(a) Find, in simplest form, $\mathrm{f}'(x)$.

(4)

(b) Hence find the range of f.

(3)